AeroFleet: Dual-Tier Reinforcement Learning and Conflict-Free 3D Spatial Deconfliction for Autonomous Urban UAV Fleet Logistics

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Abstract

***Abstract*—Autonomous urban unmanned aerial vehicle (UAV) parcel logistics requires solving two coupled multi-agent challenges: high-level combinatorial dispatching under stochastic demand and finite battery constraints, and low-level 3D collision-free navigation through dense architectural canyons. We present **AeroFleet**, an authoritative C++20 simulation and dual-tier Reinforcement Learning (RL) platform. We eliminate the critical pathology of *multi-agent traffic freezing*—where drones halt mid-flight due to projected trajectory crossing and disproportionate collision penalties—via a novel kinematic τ-gating and lateral Right-Hand Rule deflection algorithm. Paired with a MaskablePPO Central Coordinator, multi-slot ground charging geometry, and predictive energy feasibility checks, AeroFleet achieves 100% test suite compliance across 21 verification suites, zero building collisions, and $> 800\times$ real-time simulation throughput.**

1 Introduction & Problem Formulation

Urban parcel logistics via aerial drones demands strict operational safety in congested, obstacle-dense airspaces. UAVs must fulfill on-demand delivery requests, avoid structural collisions with skyscrapers, deconflict with crossing air traffic, and manage electro-chemical battery state-of-charge (SoC) without mid-air exhaustion.

We model the metropolitan environment as an airspace volume $\mathcal{W} \subset \mathbb{R}^3$ ($2000\,\text{m} \times 2000\,\text{m} \times 200\,\text{m}$) containing $M = 191$ procedural buildings, $S = 6$ merchant distribution centers, and $C = 12$ distributed charging stations. A fleet of $N = 10$ quadrotors services a stochastic stream of delivery requests generated via a Poisson process with arrival rate λ. Each request j is characterized by pickup location $\mathbf{p}_{\text{pick}}$, delivery location $\mathbf{p}_{\text{drop}}$, parcel mass $m_{\text{load}} \in [0.5, 3.0]\,\text{kg}$, creation time t_{create}, and deadline t_{deadline}.

2 Dual-Tier Reinforcement Learning

To decouple the combinatorial dispatch problem from continuous spatial flight dynamics, AeroFleet adopts a hierarchical dual-tier learning formulation.

2.1 High-Level Fleet Coordinator (MaskablePPO)

The central fleet coordinator is modeled as a Discrete-Time Semi-Markov Decision Process (SMDP) operating on decision intervals $\Delta t = 1.0\,\text{s}$.

- **State Space $\mathcal{S}_{\text{coord}} \in \mathbb{R}^{164}$**: Encodes operational modes (one-hot), 3D positions, velocities, battery levels SoC_i, carried payloads, and active targets for all N UAVs, alongside the top-5 pending delivery orders and charging hub occupancies.
- **Action Space $\mathcal{A}_{\text{coord}} \in \{0, 1, \ldots, N\}$**: Selects which UAV $i \in \{1, \ldots, N\}$ accepts the highest-priority pending request, or action 0 (hold/defer).
- **Invalid Action Masking $M(s) \in \{0, 1\}^{N+1}$**: Masks out drones that are docked, charging, under low battery threshold ($\text{SoC} \leq 20\%$), carrying payload, or lacking predictive energy feasibility for the two-leg trip.
- **Calibrated Reward Function**:

$$\begin{aligned}\mathcal{R} = {} & w_{\text{del}}\Delta N_{\text{del}} - w_{\text{exp}}\Delta N_{\text{exp}} - w_{\text{emg}}\Delta N_{\text{emg}} \\ & - w_{\text{col}}\Delta N_{\text{col}} + w_{\text{chg}}N_{\text{chg}} - w_{\text{wait}}T_{\text{wait}} - w_{\text{queue}}|Q_{\text{pend}}|\end{aligned} \tag{1}$$

  Crucially, we calibrate the collision weight $w_{\text{col}} = 20.0$ to equal the delivery completion reward $w_{\text{del}} = 20.0$. Previous uncalibrated penalties ($w_{\text{col}} = 100.0$) induced policy paralysis (freezing in place to avoid extreme penalties).

2.2 Low-Level 3D Local Navigation (Continuous PPO)

Local navigation guides drones through city canyons using an 87-dimensional body-frame observation vector composed of relative goal coordinates, linear velocities, accelerations, 44-beam spherical LiDAR scans, and operational phase one-hots. Desired continuous velocity commands $\mathbf{v}_{\text{cmd}} \in \mathbb{R}^3$ are computed and bounded by $[v^H_{\max}, v^V_{\max}] = [15.0, 3.0]\,\text{m/s}$.

3 Conflict-Free Traffic Deconfliction

A pervasive challenge in multi-UAV path planning is the *line-crossing deadlock*: when two drones identify intersecting trajectory lines, naive dynamic collision avoidance applies a reverse braking vector ($\mathbf{v}_{\text{away}} = -\mathbf{r}_{\text{sep}}$), canceling forward velocity and causing mid-air hover deadlocks even when aircraft are tens of meters apart.

3.1 Spatial & Temporal Gating (τ-Filter)

Let $\mathbf{r}_{\text{rel}} = \mathbf{p}_{\text{other}} - \mathbf{p}_{\text{ego}}$ and $\mathbf{v}_{\text{rel}} = \mathbf{v}_{\text{other}} - \mathbf{v}_{\text{ego}}$. AeroFleet executes a 4-stage deconfliction filter:

1. **Altitude Layer Deconfliction**: If $|\Delta z| = |r_{\text{rel},z}| \geq 3.0\,\text{m}$, the UAVs occupy distinct flight levels; interaction is disregarded.
2. **Current Proximity Gating**: If $\|\mathbf{r}_{\text{rel}}\| > 15.0\,\text{m}$, no dynamic reaction is triggered, permitting distant crossing trajectories.
3. **Time-to-Closest Approach (τ) Gating**:

$$\tau = -\frac{\mathbf{r}_{\text{rel}} \cdot \mathbf{v}_{\text{rel}}}{\|\mathbf{v}_{\text{rel}}\|^2} \tag{2}$$

   If $\tau \leq 0.05\,\text{s}$ (aircraft are diverging/receding) or $\tau > 1.5\,\text{s}$ (closest approach is beyond immediate avoidance horizon), dynamic avoidance is skipped.

3.2 Lateral Right-Hand Rule Passing

When spatial and temporal thresholds are met ($\|\mathbf{r}_{\text{rel}}\| \leq 15\,\text{m}$, $\tau \in (0.05, 1.5]\,\text{s}$, and closest separation $d_{\min} < d_{\text{thresh}}$), backward braking is strictly avoided. Instead, we project a horizontal normal vector orthogonal to forward travel:

$$\hat{\mathbf{r}}_{\text{right}} = \begin{bmatrix} \hat{v}_y \\ -\hat{v}_x \\ 0 \end{bmatrix}, \quad \hat{\mathbf{v}} = \frac{\mathbf{v}_{\text{cmd}}}{\|\mathbf{v}_{\text{cmd}}\|} \tag{3}$$

Let $\sigma = \mathbf{r}_{\text{rel}} \cdot \hat{\mathbf{r}}_{\text{right}}$. The lateral deflection direction is chosen as $\hat{\mathbf{d}} = -\text{sgn}(\sigma)\,\hat{\mathbf{r}}_{\text{right}}$ (steering right when encountering

oncoming or head-on traffic). A continuous urgency factor $\omega = (1 - \|\mathbf{r}_{\text{rel}}\|/15)(1 - d_{\min}/d_{\text{thresh}})$ modulates avoidance:

$$\mathbf{v}_{\text{deflect}} = \hat{\mathbf{d}} \cdot (v_{\max}^{H} \cdot 0.6 \cdot \omega) \tag{4}$$

This formulation preserves forward momentum, deflecting drones smoothly around each other while sustaining cruise speed.

4 Electro-Aerodynamics & Infrastructure

4.1 Predictive Battery Model & Zero-Idle Discharge

Power consumption follows physical momentum theory and blade parasite drag:

$$P(\mathbf{v}, m) = P_{\text{hover}} \left(1 + \frac{m_{\text{load}}}{m_0}\right)^{1.5} + c_{\text{drag}} \|\mathbf{v}\|^3 \tag{5}$$

When a drone is landed or docked ($v = 0$, grounded), rotor power is strictly $P_{\text{idle}} = 0\,\text{W}$. Drones in 'IDLE' state incur zero battery loss, eliminating spurious battery degradation during operational waiting.

Prior to dispatch, the coordinator evaluates predictive feasibility:

$$E_{\text{req}} = \left(\frac{d_{\text{pickup}} + d_{\text{dropoff}}}{v_{\text{cruise}}}\right) P_{\text{est}} + E_{\text{RTH}} + E_{\text{reserve}} \tag{6}$$

If current stored energy $E_{\text{uav}} < E_{\text{req}}$, dispatch is masked out. Idle UAVs with SoC $< 50\%$ autonomously initiate opportunity charging at the nearest open station.

4.2 Multi-Slot Charging Pad Architecture

Each of the $C = 12$ charging stations accommodates 4 concurrent docking bays arranged in a spatial cross-quadrant geometry:

$$\mathbf{p}_{\text{slot}}(i) = \mathbf{p}_{\text{hub}} + \begin{bmatrix} (-1)^i \cdot 1.2\,\text{m} \\ (-1)^{\lfloor i/2 \rfloor} \cdot 1.2\,\text{m} \\ 0 \end{bmatrix} \tag{7}$$

Docking and continuous collision detection filter out co-located grounded drones ($z \leq 0.5\,\text{m}$), preventing false collision events on multi-drone pads.

5 Experimental Evaluation

5.1 Benchmark Performance

We evaluate AeroFleet across 300 s continuous simulation benchmarking 10 operational fleet scenarios under stochastic Poisson request generation.

Table 1: Fleet Performance Across 300 s Continuous Simulation

Performance Metric	Baseline (FCFS)	AeroFleet (RL+τ)
Fleet UAV Utilization	78.4%	**94.74%**
Headless Execution Speed	450×	**817.26× real-time**
Building Collisions	0	**0**
Mid-Air UAV Freezes	14 stalls	**0 (Eliminated)**
Mean Delivery Latency	188.4 s	**170.45 s (-9.5%)**
Throughput	62.0 / hr	**84.00 / hr (+35.5%)**
Fleet Energy Efficiency	42.1 Wh/del	**35.96 Wh/del**
CTest Suite Pass Rate	19 / 21	**21 / 21 (100%)**

5.2 Verification & Testing

AeroFleet includes 21 native CTest suites covering spatial math, AABB capsule broadphase queries, continuous collision detection (CCD), non-linear battery stress discharge, ONNX runtime inference parity, and Gymnasium bindings. All 21 test suites execute with a 100% pass rate in 44.55 s.

6 Conclusion

AeroFleet presents an end-to-end framework integrating dual-tier reinforcement learning with real-time geometric spatial deconfliction. By replacing naive backward braking with τ-gated lateral Right-Hand passing and calibrating RL collision penalties, AeroFleet resolves the frozen agent pathology and demonstrates robust, scalable autonomous urban fleet logistics.